

Sara RJ Wilson

📍 East Lansing, MI ✉ srjw28@gmail.com 🌐 Webpage in LinkedIn 🐙 GitHub

Education

Michigan State University Ph.D. Computational Mathematics, Science, and Engineering <i>Advised by Prof. Andrew Christlieb & Prof. Brian O'Shea</i>	East Lansing, MI Expected May 2031
University of Pittsburgh B.S. Mathematics <i>Cum Laude</i>	Pittsburgh, PA May 2026

Honors & Awards

- Edna M. Heck Scholarship
- Society of Women Engineers Certificate of Merit in Math and Science

Skills

Programming: Python (NumPy, SciPy, Pandas, Matplotlib), C++, MATLAB, Java
Modeling: Parameter recovery (WENDy), sparse/regularized regression (Ridge, LASSO, STLSQ), system discovery (SINDy, WSINDy), numerical linear algebra, optimization
Numerical Methods: Finite differences, finite element methods, PDE solvers, Riemann solvers
Graduate Coursework (2025 – 2027): Advanced Scientific Computing, Numerical Differential Equations, Numerical Linear Algebra, Physical Electronics, Parallel Computing, Data Science, Electrodynamics of Plasmas

Research Experience

Michigan State University <i>Graduate Student Researcher</i>	East Lansing, MI Jun 2026 – Current
◦ Developing Weak-form Estimation of Nonlinear Dynamics (WENDy) algorithms to recover model parameters from noisy data, developing statistical and machine-learning methods for inverse problems.	
Semiconductor Research Corporation <i>Research Scholar</i>	Tuscaloosa, AL Jun – Jul 2025
◦ Modeled and simulated wireless power transfer coil geometries, quantifying trade-offs between inductive coupling and system efficiency across design parameters.	
West Virginia University <i>Research Assistant</i>	Morgantown, WV Jun – Jul 2024
◦ Built and analyzed finite difference schemes to model a 2×2 hyperbolic balance law system with time-dependent wave speeds; validated numerical stability and convergence against analytical benchmarks.	

Publications

J. Frew, N. Keyser, E. Kim, G. Paddock, C. Tumbleston, **S. Wilson**, C. Tsikkou, “An Analysis of a 2×2 Keyfitz–Kranzer Type Balance System with Varying Generalized Chaplygin Gas,” *Physics of Fluids*, Sept 2024. DOI: 10.1063/5.0231413

Talks & Presentations

Directed Reading Program, Carnegie Mellon University – *Computational Plasma Physics*
Summer Research Symposium, University of Alabama – *Wireless Power Transfer*
Pi Mu Epsilon Poster Session, Joint Mathematics Meetings – *Numerical Analysis of the Riemann Problem for a Cosmological 2×2 Balance System*
Summer Research Symposium, West Virginia University – *Numerical Analysis of the Riemann Problem for a Cosmological 2×2 Balance System*

Additional Training

Directed Reading Program, Carnegie Mellon University – *Computational Plasmas Physics*
Summer School, Princeton University – *Fluids & Computer Assisted Proofs*
Summer School, Texas A&M University – *Modeling & Simulation of Partial Differential Equations*

Teaching & Leadership

- **Mathematics Tutor & Teaching Assistant**, University of Pittsburgh (2023–2026) – Algebra, Calculus I–III, Linear Algebra, Differential Equations, Numerical Analysis; communicated quantitative concepts clearly to diverse audiences.
- **REU Mentor**, West Virginia University – Trained incoming researchers in MATLAB and numerical methods; work contributed to acknowledgments in two publications.